

Technology News

Robots of the future

In the next 40 years, the number of people aged 65 and over is set to increase by two billion worldwide. With that ageing will come the need to help families and other caregivers cope with decreased mobility and chronic diseases such as Parkinson's and Alzheimer's.

Tandy Trower, the general manager of Microsoft Robotics, says that robots could have a 'profound impact' when it comes to helping seniors communicate with family members and each other, or reminding someone to take their medications. Even just having robots do lightweight transport of

Robots are, of course, also making an impact in warfare. In Iraq, the US military is using robots to help defuse explosive devices. In Afghanistan, tele-operated robots, not soldiers, are being sent into caves to scout for enemies. Remote-controlled armed robots are already in the pipeline. Qinetiq, which supplies a line of robots to the US military and law enforcement agencies, has refit its TALON model with a weapon.

A gadget that watches how you drive

Using gadgets while you're driving can be a great distraction, but using a gadget that watches you while you're driving can be a very good thing.

According to John D. Lee, a professor of mechanical and industrial engineering at the University of Iowa, more than 40,000 people die every year in motor vehicle crashes,

and research indicates that failures of attention—including distractions or drowsiness—probably played a role in most of those crashes. Crashes and near-crashes are about three times as likely to happen when the driver is performing a complex task not related to driving (such as dialing a phone or even texting), and twice as likely during a moderately complex task (such as inserting a CD).

So how does watching the driver help? Lee installed a special camera system that saves video snippets for the ten seconds before and after every abrupt movement on the road. After the feedback sessions, the number of events triggered by risky drivers declined 89 per cent, and the

rate of risky driving remained low even six weeks later.

After-the-fact monitoring systems are being used to check up on motor-fleet drivers as well as teenagers, and it won't be long before real-time monitors show up as well. There are video cameras that are actually being put into high-end Toyota and Volvo cars. These video-based face-tracking systems tell whether the driver is looking at the road or looking into the car.

Such a feature, sometimes known as a 'drowsiness warning system,' may be provided as an optional package with other advanced safety features such as forward-collision avoidance, side-collision warning and blindspot detection. Japan's Nissan Motor Co. even suggests that future cars could automatically sound an alarm and release 'a stimulating mint fragrance' if they sense that the driver is dozing off.

Mind-reading headsets

For people suffering from disabilities, here is some good news. Researchers



Robosoft's robuLAB10 robot that follows and assists elderly persons in their own home

objects from one room to another could be tremendously valuable.

This is not as far away as you might think, judging by some of the robots exhibited recently at the *Robo Business 2009 Conference and Expo* in Boston.

For instance, Robosoft, a French company, showed off robuLAB10—a robot that follows and assists elderly persons in their own home. It comes equipped with a touch-screen computer, a voice interface and special robot navigation software developed by Silicon Valley's SRI International.



Emotiv helmet